Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (38 marks). Same format as the final exam.

Answer Part A by circling one option. Answer Part B in full, showing your working. The answer key is at the back.

Part A · Multiple choice

Choose one answer for each question.

A1 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO1

The IUPAC name of $CH_3CH_2CH_2COOH$ is:

A. butanoic acid
B. propanoic acid
C. butanal
D. 1-butanol

A2 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO1

In the common name α-chlorobutyric acid, the Cl is on:

A. C2, the carbon next to –COOH
B. C1, the carboxyl carbon
C. C3
D. C4

A3 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO2

Which reaction of a carboxylic acid produces CO_2 gas?

A. with $NaHCO_3$
B. with NaOH
C. with ethanol and H^+
D. with water

A4 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO3

The ester formed from ethanoic acid and 1-propanol is:

A. propyl ethanoate
B. ethyl propanoate
C. propyl propanoate
D. ethyl ethanoate

A5 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO3

Saponification is:

A. basic hydrolysis of an ester
B. acidic hydrolysis of an amide
C. the formation of an ester
D. oxidation of an alcohol

A6 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO3

Which is an ester of an inorganic acid used to relieve angina?

A. glyceryl trinitrate
B. aspirin
C. ethyl acetate
D. methyl salicylate

A7 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO4

$(CH_3CH_2)_3N$ is:

A. a tertiary amine
B. a secondary amine
C. a primary amine
D. an amide

A8 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO4

Methylamine reacts with HCl to form:

A. methylammonium chloride
B. methyl chloride + NH_3
C. N-methylformamide
D. chloromethane

A9 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO4

Tertiary amines boil lower than primary amines of the same molar mass because they:

A. have no N–H, so cannot hydrogen bond to each other
B. are ionic
C. are heavier
D. contain oxygen

A10 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO5

The functional group in paracetamol that links the ring to the acetyl group is:

A. an amide
B. an ester
C. an amine
D. a ketone

A11 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO5

Which cannot form an amide with acetic acid?

A. trimethylamine
B. ammonia
C. methylamine
D. dimethylamine

A12 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO5

Nylon-66 is a:

A. polyamide made from a diacid and a diamine
B. polyester made from a diacid and a diol
C. addition polymer of ethene
D. natural protein

Part B · Structured questions

Write your answers on paper, then self-mark with the worked solution.

Question B1 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO2 8 marks, C3

(a) Give the IUPAC names of (2 marks)

(i) $(CH_3)_2CHCH_2COOH$

(ii) $BrCH_2CH_2COOH$.

(b) Acetic acid (60 g/mol) boils at 118 °C, but 1-propanol (60 g/mol) boils at 97 °C. Explain. (2 marks)

(c) Chloroacetic acid has pK_a 2.85 and acetic acid has pK_a 4.74. State which is the stronger acid and explain why. (2 marks)

(d) Write an equation for the reaction of benzoic acid with aqueous sodium bicarbonate and name the organic product. (2 marks)

Question B2 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO3 8 marks, C3

Ethyl butanoate gives pineapple flavour to syrups.

(a) Write an equation for its preparation from a carboxylic acid and an alcohol, naming both reactants and stating the conditions. (3 marks)

(b) Give two ways to increase the yield of the ester. (2 marks)

(c) Write an equation for the hydrolysis of ethyl butanoate with sodium hydroxide and name both products. (3 marks)

Question B3 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO4 8 marks, C3

(a) Name and classify (1°, 2° or 3°) each amine: (3 marks)

(i) $CH_3CH_2NHCH_3$

(ii) $(CH_3CH_2)_3N$

(iii) $(CH_3)_2CHNH_2$.

(b) Propylamine and trimethylamine are both C_3H_9N. Propylamine boils at 48 °C; trimethylamine boils at 3 °C. Explain. (2 marks)

(c) Write an equation for the reaction of methylamine with hydrochloric acid and name the product. (2 marks)

(d) Give one reason why amine drugs are formulated as hydrochloride salts. (1 mark)

Question B4 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO5 8 marks, C3

(a) Write an equation for the reaction of acetic acid with ethylamine on heating, and name the organic product. (2 marks)

(b) Acetamide melts at 82 °C, but is not basic, while ethylamine is a base. Explain both facts. (2 marks)

(c) Give the products of hydrolysing propanamide (2 marks)

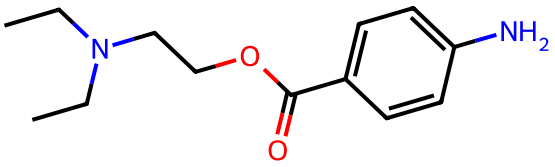
(i) in acid

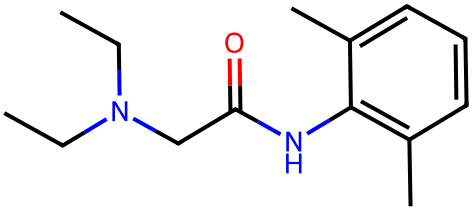
(ii) in sodium hydroxide.

(d) Explain why trimethylamine does not form an amide with acetic acid, and say what forms instead. (2 marks)

Question B5 Topic: Carboxylic Acid, Ester, Amines, Amide | CH4-LO5 | 6 marks, C4

Procaine (top) and lidocaine (bottom) are local anaesthetics.





(a) Identify the functional group that links the two halves of each molecule. (2 marks)

(b) Procaine acts for a short time, lidocaine for longer. Explain using your answer to (a). (2 marks)

(c) Both are injected as hydrochloride salts. Identify the group that forms the salt, and explain why the salt is used. (2 marks)

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PHD125 Pharmaceutical Organic Chemistry · Carboxylic Acid, Ester, Amines, Amide

Tutorial: exam-style practice

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Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (38 marks). Same format as the final exam.

Answer key

Part A · Multiple choice

A1 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO1

The IUPAC name of $CH_3CH_2CH_2COOH$ is:

A. butanoic acid

B. propanoic acid

C. butanal

D. 1-butanol

Four carbons, including the carboxyl carbon.

A2 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO1

In the common name α-chlorobutyric acid, the Cl is on:

A. C2, the carbon next to –COOH

B. C1, the carboxyl carbon

C. C3

D. C4

Greek letters start next to the carboxyl carbon (Ball 15.1).

A3 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO2

Which reaction of a carboxylic acid produces CO_2 gas?

A. with $NaHCO_3$

B. with NaOH

C. with ethanol and H^+

D. with water

Carbonates and bicarbonates release CO_2.

A4 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO3

The ester formed from ethanoic acid and 1-propanol is:

A. propyl ethanoate

B. ethyl propanoate

C. propyl propanoate

D. ethyl ethanoate

Alcohol part (propyl) first, then the acid part (-ate).

A5 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO3

Saponification is:

A. basic hydrolysis of an ester

B. acidic hydrolysis of an amide

C. the formation of an ester

D. oxidation of an alcohol

It gives a carboxylate salt (soap from fats).

A6 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO3

Which is an ester of an inorganic acid used to relieve angina?

A. glyceryl trinitrate

B. aspirin

C. ethyl acetate

D. methyl salicylate

An ester of glycerol and nitric acid (FGOB 17.6).

A7 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO4

$(CH_3CH_2)_3N$ is:

A. a tertiary amine

B. a secondary amine

C. a primary amine

D. an amide

Three ethyl groups on N.

A8 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO4

Methylamine reacts with HCl to form:

A. methylammonium chloride

B. methyl chloride + NH_3

C. N-methylformamide

D. chloromethane

$CH_3NH_2 + HCl \rightarrow CH_3NH_3^+Cl^-$.

A9 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO4

Tertiary amines boil lower than primary amines of the same molar mass because they:

A. have no N–H, so cannot hydrogen bond to each other

B. are ionic

C. are heavier

D. contain oxygen

Their boiling points are similar to ethers (FGOB 16.3).

A10 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO5

The functional group in paracetamol that links the ring to the acetyl group is:

A. an amide

B. an ester

C. an amine

D. a ketone

N-(4-hydroxyphenyl)acetamide.

A11 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO5

Which cannot form an amide with acetic acid?

A. trimethylamine

B. ammonia

C. methylamine

D. dimethylamine

A 3° amine has no H on N; it only forms a salt (FGOB 17.3).

A12 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO5

Nylon-66 is a:

A. polyamide made from a diacid and a diamine

B. polyester made from a diacid and a diol

C. addition polymer of ethene

D. natural protein

Adipic acid + hexanediamine (FGOB 17.5).

Part B · Structured questions

Question B1 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO2 8 marks, C3

(a) Give the IUPAC names of (2 marks)

(i) $(CH_3)_2CHCH_2COOH$

(ii) $BrCH_2CH_2COOH$.

(b) Acetic acid (60 g/mol) boils at 118 °C, but 1-propanol (60 g/mol) boils at 97 °C. Explain. (2 marks)

(c) Chloroacetic acid has pK_a 2.85 and acetic acid has pK_a 4.74. State which is the stronger acid and explain why. (2 marks)

(d) Write an equation for the reaction of benzoic acid with aqueous sodium bicarbonate and name the organic product. (2 marks)

(a) **(i)** 3-methylbutanoic acid

(ii) 3-bromopropanoic acid.

(b) Both form hydrogen bonds, but two acid molecules are held together by two hydrogen bonds (a dimer), using both the C=O and the O–H, so more energy is needed to separate them.

(c) Chloroacetic acid is stronger (smaller pK_a): the electronegative Cl pulls electron density away and stabilises the carboxylate ion.

(d) $C_6H_5COOH + NaHCO_3 \rightarrow C_6H_5COO^-Na^+ + H_2O + CO_2$; sodium benzoate.

Step	Marking criteria	Marks
1	(i) 3-methylbutanoic acid	1
2	(ii) 3-bromopropanoic acid	1
3	Acids form hydrogen-bonded dimers (two H-bonds)	1
4	More energy needed than for the alcohol	1
5	Chloroacetic acid is stronger (smaller pK_a)	1
6	Cl withdraws electrons, stabilising the anion	1
7	Correct balanced equation with CO_2	1
8	Sodium benzoate	1

	Total	**8**

Question B2 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO3 8 marks, C3

Ethyl butanoate gives pineapple flavour to syrups.

(a) Write an equation for its preparation from a carboxylic acid and an alcohol, naming both reactants and stating the conditions. (3 marks)

(b) Give two ways to increase the yield of the ester. (2 marks)

(c) Write an equation for the hydrolysis of ethyl butanoate with sodium hydroxide and name both products. (3 marks)

(a) $CH_3CH_2CH_2COOH + CH_3CH_2OH \rightleftharpoons CH_3CH_2CH_2COOCH_2CH_3 + H_2O$; butanoic acid and ethanol; heat with a little concentrated H_2SO_4.

(b) Use excess ethanol; remove the water (or the ester) as it forms.

(c) $CH_3CH_2CH_2COOCH_2CH_3 + NaOH \rightarrow CH_3CH_2CH_2COO^-Na^+ + CH_3CH_2OH$; sodium butanoate and ethanol.

Step	Marking criteria	Marks
1	Butanoic acid and ethanol named	1
2	Correct equation with water (reversible)	1
3	Conc. H_2SO_4 catalyst, heat	1
4	Excess alcohol (or acid)	1
5	Remove water or ester	1
6	Correct equation	1
7	Sodium butanoate	1
8	Ethanol	1
	Total	**8**

Question B3 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO4 8 marks, C3

(a) Name and classify (1°, 2° or 3°) each amine: (3 marks)

- **(i)** $CH_3CH_2NHCH_3$
- **(ii)** $(CH_3CH_2)_3N$
- **(iii)** $(CH_3)_2CHNH_2$.

(b) Propylamine and trimethylamine are both C_3H_9N. Propylamine boils at 48 °C; trimethylamine boils at 3 °C. Explain. (2 marks)

(c) Write an equation for the reaction of methylamine with hydrochloric acid and name the product. (2 marks)

(d) Give one reason why amine drugs are formulated as hydrochloride salts. (1 mark)

(a) **(i)** ethylmethylamine, 2°

(ii) triethylamine, 3°

(iii) isopropylamine, 1°.

(b) Propylamine (1°) has N–H bonds and forms hydrogen bonds between its molecules; trimethylamine (3°) has no H on N, so it cannot.

(c) $CH_3NH_2 + HCl \rightarrow CH_3NH_3^+Cl^-$; methylammonium chloride.

(d) Any one: the salt is ionic and dissolves in water; it is a stable, odourless solid.

Step	Marking criteria	Marks
1	(i) ethylmethylamine, 2°	1
2	(ii) triethylamine, 3°	1
3	(iii) isopropylamine, 1°	1
4	Propylamine forms H-bonds (N–H)	1
5	Trimethylamine has no N–H	1
6	Correct equation	1
7	Methylammonium chloride	1
8	Water-soluble (or stable/odourless)	1
	Total	**8**

Question B4 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO5 8 marks, C3

(a) Write an equation for the reaction of acetic acid with ethylamine on heating, and name the organic product. (2 marks)

(b) Acetamide melts at 82 °C, but is not basic, while ethylamine is a base. Explain both facts. (2 marks)

(c) Give the products of hydrolysing propanamide (2 marks)

(i) in acid

(ii) in sodium hydroxide.

(d) Explain why trimethylamine does not form an amide with acetic acid, and say what forms instead. (2 marks)

(a) $CH_3COOH + CH_3CH_2NH_2 \rightarrow CH_3CONHCH_2CH_3 + H_2O$; N-ethylacetamide (N-ethylethanamide).

(b) Amide molecules form many hydrogen bonds (N–H···O=C), so acetamide is a solid with a high melting point. The N lone pair is pulled towards the C=O, so it does not accept H^+; in ethylamine the lone pair is free.

(c) **(i)** propanoic acid + ammonium ion (NH_4^+)

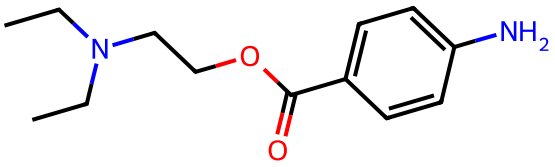
(ii) sodium propanoate + ammonia.

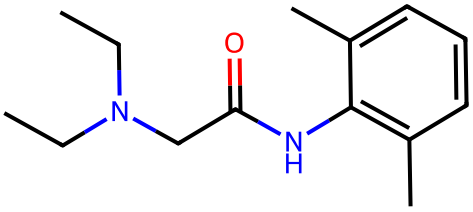
(d) A 3° amine has no H on the nitrogen to lose with the –OH as water; an acid–base reaction gives the salt trimethylammonium acetate instead.

Step	Marking criteria	Marks
1	Correct equation with water	1
2	N-ethylacetamide (N-ethylethanamide)	1
3	H-bonding network → high mp	1
4	Lone pair delocalised towards C=O → not basic	1
5	Acid: propanoic acid + NH_4^+	1
6	Base: propanoate salt + NH_3	1
7	No H on N in a 3° amine	1
8	Forms a salt (trimethylammonium acetate)	1
	Total	**8**

Question B5 Topic: Carboxylic Acid, Ester, Amines, Amide CH4-LO5 6 marks, C4

Procaine (top) and lidocaine (bottom) are local anaesthetics.





(a) Identify the functional group that links the two halves of each molecule. (2 marks)

(b) Procaine acts for a short time, lidocaine for longer. Explain using your answer to (a). (2 marks)

(c) Both are injected as hydrochloride salts. Identify the group that forms the salt, and explain why the salt is used. (2 marks)

(a) Procaine: an ester (–COO–). Lidocaine: an amide (–CONH–).

(b) Esters are hydrolysed quickly by esterase enzymes in the blood, so procaine is inactivated fast. Amides resist hydrolysis and are broken down slowly in the liver, so lidocaine lasts longer.

(c) The tertiary amine group (the N with two ethyl groups) accepts H^+ from HCl. The salt is ionic, so it dissolves in water for an injection.

Step	Marking criteria	Marks
1	Procaine: ester	1
2	Lidocaine: amide	1
3	Esters hydrolysed quickly by plasma esterases	1
4	Amides resist hydrolysis → longer action	1
5	Tertiary amine forms the salt	1
6	Salt is ionic → water-soluble injection	1

	Total	**6**

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